



High-fidelity integrated co-simulation model for dynamic analysis of onshore wind turbines with Steel–Concrete Hybrid Tower

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ABSTRACT

The Steel–Concrete Hybrid Tower (SCHT) structure is complex and exhibits significant nonlinear characteristics, presenting a longstanding challenge in dynamic modeling and analysis within the industry. In this study, a wind-rotor-tower integrated coupling model is established using the subsystem co-simulation method to facilitate the dynamic analysis of detailed structures. A multi-body model of the rotor and nacelle assembly is created using Simpack, while a nonlinear finite element model of the SCHT is developed in Abaqus. The co-simulation engine enables data exchange between the different solvers, allowing for detailed modeling of the tower structure while maintaining analysis efficiency. Therefore, a high-fidelity integrated co-simulation model is established accordingly. This approach enables the nonlinear time history analysis of onshore wind turbines with SCHT under coupled wind load effects. Comparative analyses with OpenFAST validates the effectiveness of the proposed subsystem co-simulation method. Subsequently, the proposed method is used to study the dynamic response of wind turbines with SCHT, investigating their distinct behaviors. Additionally, a novel transition section is proposed to mitigate both displacement and stress amplitude in the bolts.

1. Introduction

In light of escalating air pollution, the threat of global warming and growing apprehensions about the depletion of non-renewable natural resources, there is a heightened public focus on green energy today [1,2]. Wind energy, known for its minimal environmental impact and abundant availability, has gained widespread acceptance [3–6]. To harness greater wind energy at high altitudes, it becomes imperative to elevate the hub height of wind turbines [7]. There are three main engineering solutions of high tower, namely all-steel tower, all-concrete tower, and Steel–Concrete Hybrid Tower (SCHT). In Fig. 1(a), an onshore wind turbine featuring SCHT is illustrated, employing a concrete structure to replace the lower section of the elevated tower [8,9]. The concrete tower section comprises several stacked concrete segments and is firmly fixed to the concrete foundation. The steel tower section is consistent with the all-steel tower. The transition section, as depicted in Fig. 1(b), connects the steel and concrete tower sections. The connection is achieved as follows: The steel tower section is anchored to the upper Ring Flange (RF) of the transition section via bolts and nuts. A downward pressure provided by Post-Tensioning (PT) tendons and anchorages compresses the lower RF of the transition section tightly upon the upper surface of the concrete tower section. Additionally, the prestressing exerts a compressive force

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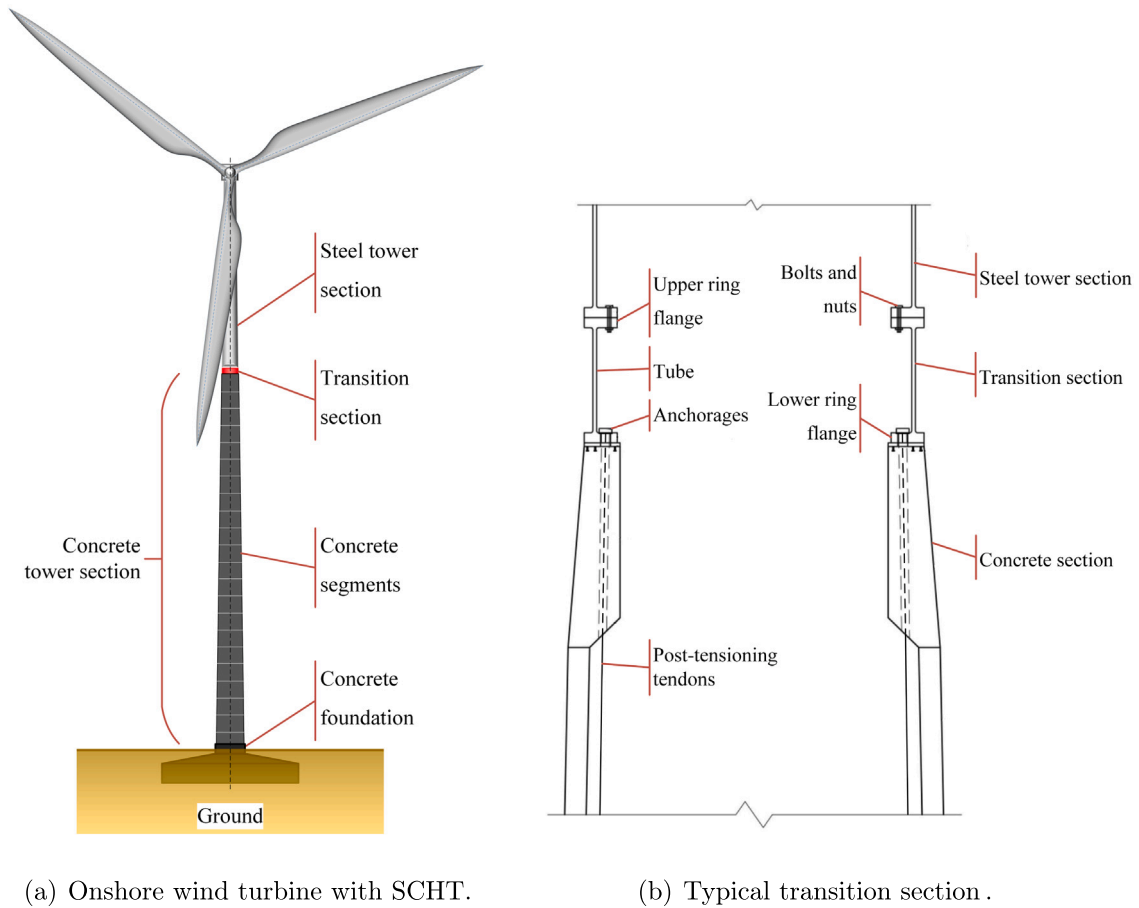


Fig. 1. Onshore Wind turbine with SCHT and its transition section (see [13]).

between the concrete tower segments [10], which prevents their separation and enhances the structural integrity of the tower. The SCHT exhibits superior structural rigidity compared to the all-steel tower, reducing susceptibility to resonance with the rotor. This inherent stability allows the SCHT to navigate complex working conditions without intricate control strategies or internal vibration damping devices [11,12]. Current research underscores the SCHT as the preferred solution for addressing elevated tower challenges. Therefore, studying the dynamic modeling and analysis of wind turbines with SCHT is essential.

In traditional wind turbine models, the modal superposition method is commonly employed for Multi-Body Dynamics (MBD) modeling to account for the elastic deformation of the all-steel tower [14]. The modal superposition method simplifies the complex dynamic behavior of structures by considering only a limited number of modes, thus making it computationally efficient for engineering applications [15]. Leveraging these advantages, mainstream aeroelastic simulation tools like OpenFAST and Flex5 employ this method to evaluate the dynamic response of wind turbine towers. Høeg [16] and Sarkar [17] developed MBD models for offshore wind turbines based on the Euler-Lagrange equations and the Kane method, respectively, combined with modal superposition method. They validated their models by comparing the results with those obtained from OpenFAST. Chen [18] analyzed the seismic response of a all-steel tower and a SCHT using modal response spectrum method.

While the modal superposition method is efficient and adequate for analyzing traditional steel tower structures with linear material models, it is inadequate for capturing the nonlinear characteristics of SCHT. The hybrid structural composition and reinforced concrete material of SCHT introduce significant nonlinearities in material properties, geometry, and contact [19]. Moreover, this method is inherently limited in its ability to provide precise stress and strain information for the structure. Therefore, the current focus of numerical modeling research for wind turbines with SCHT is primarily on the Finite Element (FE) method. Ma [20] established a multi-degree-of-freedom model and a FE model for SCHT structure and conducted comparative analyses of response spectrum and time history analyses. The research findings indicated that the seismic effects of the response spectrum method were smaller than those of the nonlinear time history analysis method. Therefore, time history analysis should be considered as a necessary supplement for seismic design of SCHT. Ercan [21] utilized the FEA software Abaqus to establish a prestressed concrete tower structure model and performed dynamic loads to assess the safety of the tower structure. De Lana [22] developed a FE model for a concrete tower and analyzed its natural frequencies, limit states, fatigue limits, etc. Using Hamilton's principle,

Huang [23] derived partial differential equations of motion for the tower, considering its variable properties such as flexural rigidity. Furthermore, Huang [24] created a cantilever beam model in OpenSees to evaluate the nonlinear response of SCHAT. From the above research, it can be seen that the FEA of SCHAT wind turbines are either conducted by applying dynamic loads to the tower alone, or by simplifying the SCHAT to beam elements and establishing an integrated model with the Rotor Nacelle Assembly (RNA). This highlights that existing studies on numerical analysis of SCHAT dynamics primarily adopt two approaches [25]: one simplifies the tower using beam or shell elements, facilitating a coupled analysis model of the wind field, rotor, and tower; the other employs solid elements to create a refined FE model of the tower but typically reduces the RNA to a point mass, neglecting the coupled effect of aerodynamic loads. Research on developing a fully integrated dynamic model that combines a high-fidelity SCHAT model with RNA remains scarce [14].

While previous studies have advanced the coupled analysis of wind turbine systems, they often ignore the nonlinear characteristics of the tower and lack attention to essential structural details, such as reinforced concrete, bolts, and prestressed tendons. As a result, current coupled dynamic analyses offer limited insights for comprehensive safety assessments of real-world scenarios. SCHAT wind turbines, in particular, present significant modeling challenges for integrated dynamic analysis. RNA analysis involves domains such as aerodynamics and control systems, while the SCHAT component demands specialized knowledge in civil and mechanical engineering, as well as nonlinear Finite Element Analysis (FEA). Besides, the entire system displays complex dynamic interactions, including both aerodynamic-structural and structural-structural coupling effects. This multidisciplinary scope, which encompasses aerodynamics, RNA control systems, civil and mechanical engineering, as well as nonlinear FEA for the SCHAT, complicates the modeling and analysis process, making it challenging to complete within a single software platform.

High-fidelity FEA often struggles with convergence and requires substantial computational time. To ensure the analysis efficiency while enabling detailed mechanical analysis of key structures, the Subsystem Co-simulation Method (SCM) is proposed and studied, which integrates Multi-Body Dynamic and Finite Element (MBD-FE) approaches [26,27] to address these challenges and limitations of traditional FE and MBD analyses. Tang [28] proposed a co-simulation method for train-track-bridge system using Simpack and OpenSees. Helsén [29] illustrated MBD modeling techniques of different levels of complexity. The research highlighted that the coupling model between the FE model and the MBD model were the most accurate. Currently, research on the SCM for wind turbines remains largely unexplored. This study proposes a novel SCM-based simulation approach tailored for SCHAT wind turbines. The model decomposes the wind turbine system into distinct subsystems, including the RNA and the tower, enabling the use of specialized software platforms to accurately simulate their unique loading conditions and dynamic behaviors. Specifically, the proposed framework integrates Simpack for MBD modeling of the RNA subsystem and Abaqus for FEA of the tower, facilitating accurate load/displacement transfer between the subsystems. This integration allows for high-fidelity coupled dynamic simulation of the wind field, rotor, and tower, capturing critical geometric, material, and boundary nonlinearities. Furthermore, the tower structure achieves refined modeling and integrated coupled nonlinear dynamic analysis. The rest of the paper is organized as follows: Section 2 focuses on the subsystem modeling of RNA and SCHAT, and introduces the modeling method for the three-dimensional fluctuating wind field. Subsequently, Section 3 introduces the co-simulation method for subsystems by establishing the MBD-FE model of a wind turbine with SCHAT. To verify accuracy of the SCM, comparative analyses with OpenFAST is conducted on the MBD-FE model of the National Renewable Energy Laboratory (NREL) 5 MW reference wind turbine. Next, Section 4 compares the MBD-FE model with the MBD and FE models, highlighting differences in dynamic responses. It further delves into the dynamic analysis of the MBD-FE model under various wind conditions. To address the issue of high bolt stress, a novel steel transition section structure is proposed to enhance the structural performance of SCHAT. Finally, Section 5 provides concluding remarks and outlines areas for further investigation. For a concise depiction of the procedure, refer to the flowchart presented in Fig. 2.

2. Model description of subsystem

This section offers a general description of an onshore wind turbine and details the modeling method for the RNA and the SCHAT subsystems. In this study, a SCHAT onshore wind turbine model is selected, and the key structural parameters are detailed in Table 1. The rotor parameters refer to the NREL 5 MW reference wind turbine [30]. The total height of the tower is 157.3 m, of which the reinforced concrete section standing at 112.0 m and the steel section measuring 45.3 m.

2.1. Modeling for RNA

2.1.1. MBD model

The MBD model of the RNA is established using Simpack, a software renowned for its capabilities in simulating the dynamics of mechanical systems. The topology of the RNA model is shown in Fig. 3. The model comprises multiple components, each represented as either a rigid or flexible body based on its role and dynamic characteristics. All components, except the blades, treated as rigid bodies. The blades are modeled as flexible bodies by defining both structural and aerodynamic properties. Structural properties include parameters such as mass distribution and stiffness, which describe the blades' physical characteristics and flexibility. Aerodynamic properties are defined by parameters like airfoil shapes and chord lengths, which determine how the blades interact with wind flow to generate lift and power. These blade properties are defined according to the parameters detailed in the technical report [32]. Rigid bodies are characterized by their mass, center of gravity, and inertia, which determine their resistance to motion and response to applied forces during operation.

The RNA model's Degrees of Freedom (DOFs) for the blades, hub, and nacelle are reduced based on linear modal and MBD formulations. The model includes three blades, each rigidly connected to a pitch flange. Each pitch flange allows one DOF relative

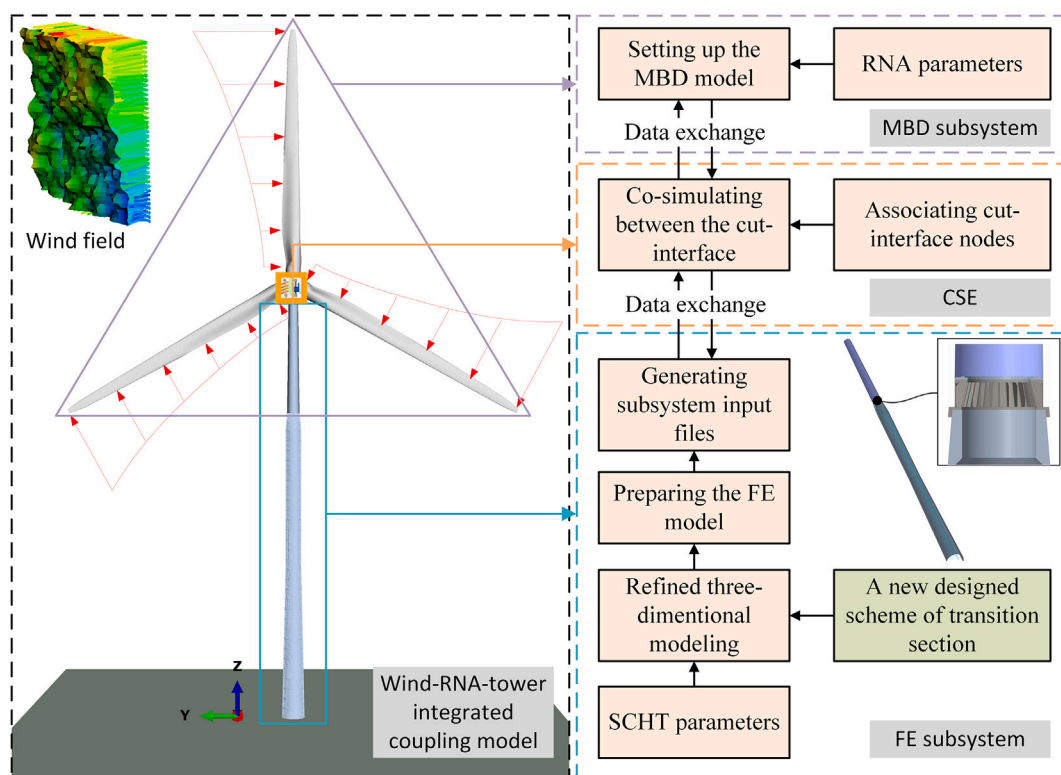


Fig. 2. Research procedure.

Table 1
Properties chosen for the SHT wind turbine.

Part	Property	Value
RNA	Rotor orientation, configuration	Upwind, 3 blades
	Rotor diameter (m)	126
	Hub height (m)	160
	Rated rotor speed (rpm)	12.1
	Rated generator speed (rpm)	1173.7
	Rotor Mass (kg)	110,000
	Nacelle Mass (kg)	240,000
Concrete tower section	Height (m)	112.0
	Segment number (pcs)	31
	Top diameter (m)	4.7
	Bottom diameter (m)	8.3
	Material	C70
	Structural-damping ratio	5% [31]
Transition section	Height (m)	1.5
	Material	Q345
Steel tower section	Height (m)	43.8
	Top diameter (m)	3.4
	Bottom diameter (m)	4.5
	Structural-damping ratio	2% [31]
	Material	Q345
PT tendons	Number	36
	Nominal cross-sectional area (mm ²)	140.0
	Material	Steel strand wire
Bolts and nuts	Thread specification	M45
	Property class	Grade 10.9

to the hub, enabling the blades to pitch. This rotation is essential for controlling the aerodynamic loads on the blades and managing the power output of the wind turbine. The hub is rigidly connected to the low-speed shaft, which in turn is flexibly coupled to the

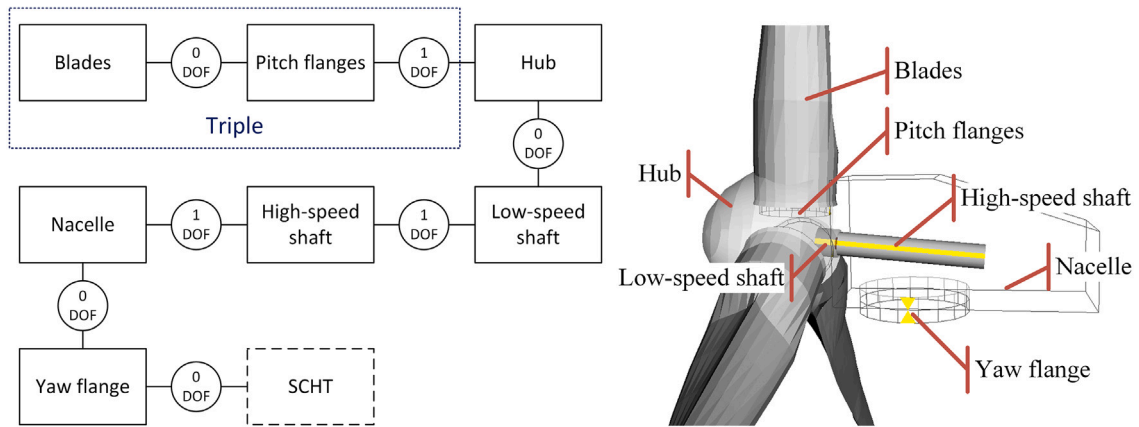


Fig. 3. Topology of RNA Model.

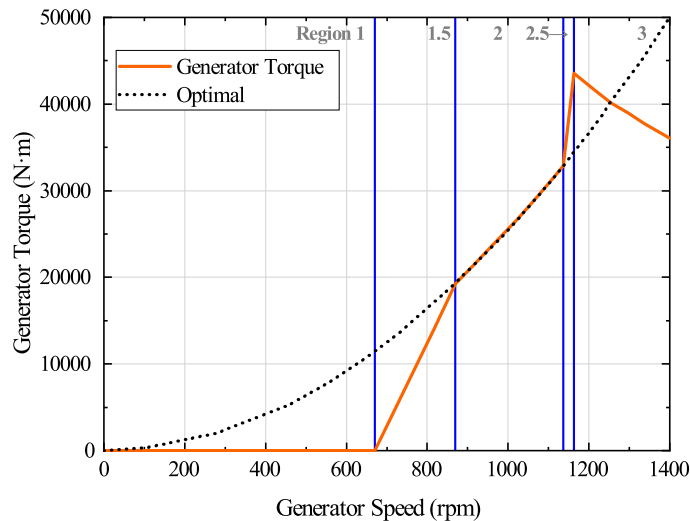


Fig. 4. Torque control curve.

high-speed shaft. This flexible connection is included to account for compliance and elasticity in the drivetrain, particularly between the hub and the generator. The high-speed shaft has rotational DOFs relative to the nacelle. Finally, the nacelle is fixed to the yaw flange.

2.1.2. Control system properties

The study also considers the generator unit's servo control system, including the blade pitch control and generator torque control configurations, to accurately replicate the operational conditions of wind turbines. The control strategy aligns with the NREL 5 MW baseline wind turbine. This section provides a concise overview of the control configuration. For detailed modeling procedures, interested readers can refer to the technical report [30].

The generator torque controller aims to maximize power capture below the rated operation point [33], operating in different control regions based on the generator speed. To achieve maximum wind energy capture, the pitch angle should be set to zero during low wind speed periods, producing the maximum aerodynamic torque for the given wind speed conditions. Fig. 4 shows the generator speed-torque response curve. In Region 1, before cut-in wind speed, the generator torque is zero, and no power is extracted from the wind. Region 2 is dedicated to optimizing power capture, where the generator torque is proportional to the square of the generator speed to maintain an optimal tip-speed ratio. Regions 1.5 and 2.5 are linear transition regions. In Region 3, when the wind speed exceeds the rated wind speed, the system activates blade pitch control. The generator power remains constant, making the generator torque inversely proportional to the generator speed. Pitch control actively decreases the pitch angle to reduce the angle of attack, thereby decreasing the blades' ability to capture wind and reducing aerodynamic loads. This regulates the generator speed and controls the power output. The pitch control employs a gain-scheduled Proportional-Integral (PI) control strategy based on the deviation of the instantaneous rotor speed from the rated speed, with a pitch range set from 0 to -90 degrees.

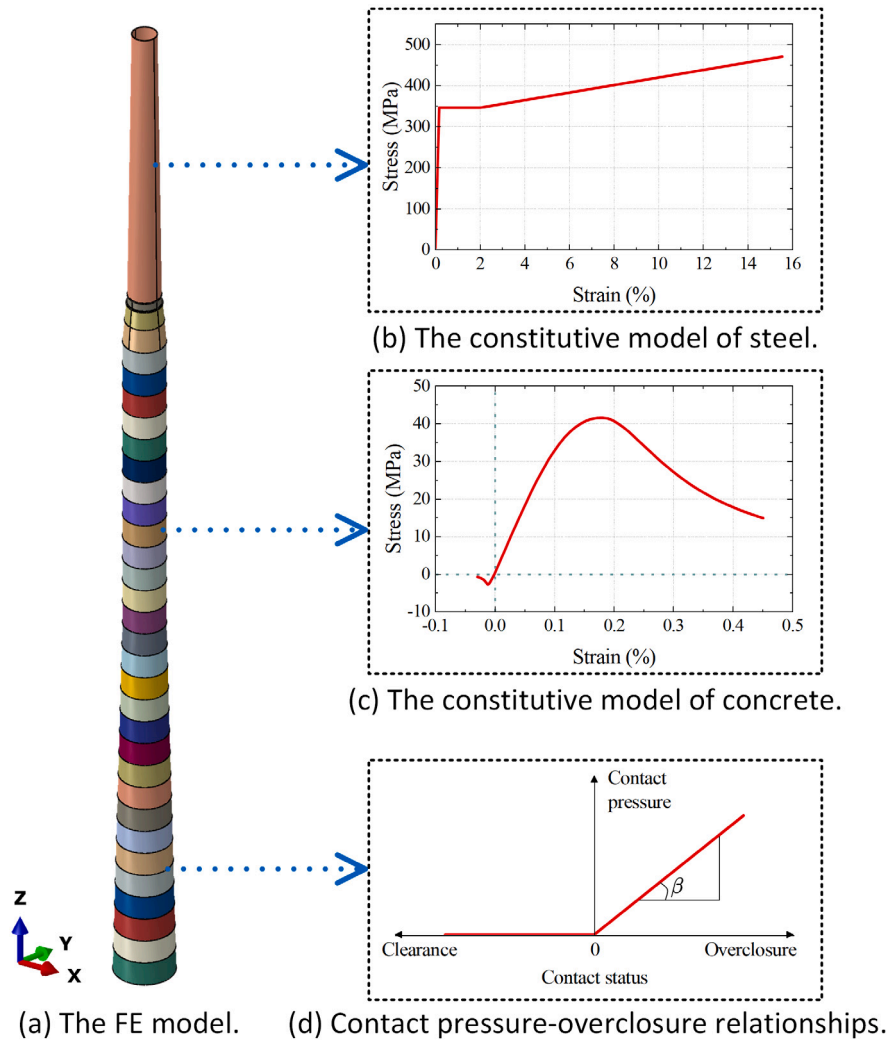


Fig. 5. The FE model in Abaqus.

2.2. Modeling for SCHAT

2.2.1. FE model

The nonlinear FE model of the SCHAT structure is developed using Abaqus, a robust software suite known for its detailed analysis of stress and strain in complex structures. This model incorporates crucial factors such as material properties and geometric configurations. To simplify the analysis and reduce computational complexity, auxiliary components that have minimal impact on the tower's dynamic characteristics are omitted. Nonetheless, the model retains a high level of detail. It includes a concrete tower section with embedded reinforcing rebars, a steel tower section, PT tendons, and a transition section featuring bolted RF connections. Initially, a detailed three-dimensional model of the SCHAT is created using Creo, a software specialized in handling complex structural designs, including intricate reinforcement details. This three-dimensional model is then imported into Abaqus for FEA, allowing for an in-depth examination of the tower's structural behavior under various conditions.

The analysis is conducted using the International System of Units (SI), employing units such as meters (m), newtons (N), kilograms (kg), seconds (s), pascals (Pa) (N/m^2), kilograms per cubic meter (kg/m^3), and meters per second squared (m/s^2). The coordinate system is defined according to the right-hand rule, as depicted in Fig. 5(a). The concrete tower section is divided into 31 distinct segments, each modeled independently. The concrete is specified as grade C70, while HRB400 steel is used for the reinforcing rebars and steel strand wire for the PT tendons. Prestressing is applied using a predefined field with a set load value of 1280 MPa. For the transition and steel tower sections, Q345 steel is specifically utilized. Table 2 summarizes the main material properties and their characteristic values employed in this study [34]. The material's constitutive models are illustrated in Figs. 5(b) and 5(c).

Hexahedral solid elements C3D8R are employed for modeling the concrete, bolts, and nuts. Reinforcing rebars and PT tendons are represented using three-dimensional truss elements T3D2 [11], while the steel tower section is modeled with shell elements

Table 2
Material properties.

Material	Density (kg/m ³)	Young's modulus (MPa)	Poisson's ratio	Compressive strength (MPa)	Yield strength (MPa)	Tensile strength (MPa)
Concrete	2400	3.70×10^4	0.2	44.5	/	2.99
Rebar	7850	2.00×10^5	0.3	/	400	540
Steel strand wire	7850	1.95×10^5	0.3	/	/	1860
Steel	7850	2.06×10^5	0.3	/	345	/

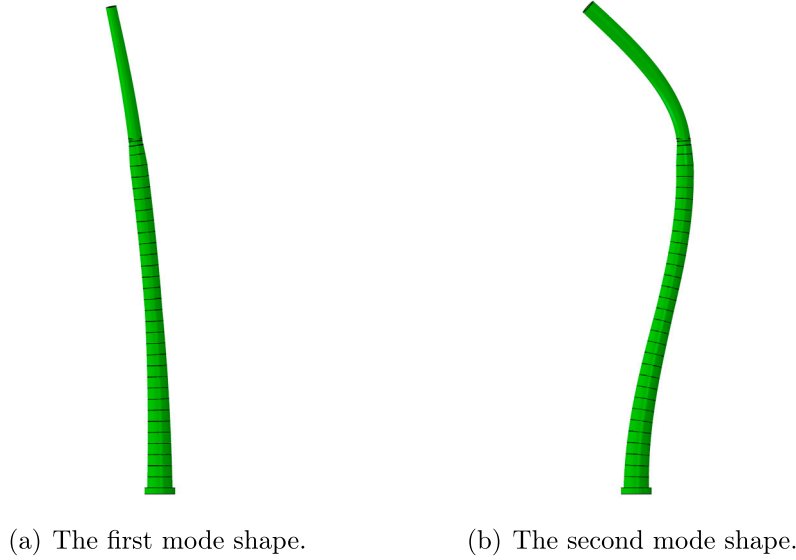


Fig. 6. The first and second mode frequencies are 0.2230 Hz and 0.8123 Hz (FE model) compared to 0.2377 Hz and 0.8434 Hz (true model), respectively.

S4R. Contact interactions between concrete segments are defined using surface-to-surface contact, with Coulomb friction applied. The friction coefficient β is set to 0.4, as shown in Fig. 5(d).

2.2.2. Modal analysis validation

This section evaluates the modal characteristics of the SCHT to assess the validity of the FE model. A global gravity load is initially applied in the negative Z -axis direction. The first two vibration modes and their corresponding natural frequencies are shown in Fig. 6. The results indicate that the natural frequencies predicted by the FE model closely match those obtained from true experimental measurements, demonstrating the accuracy and reliability of the modeling approach.

2.3. Wind field model

Accurate wind field modeling is essential for effective wind turbine design, significantly influencing the structural performance of the turbines [35]. The blades, which are designed to harness wind energy, experience the majority of wind-induced loads and are responsible for transferring these forces to the rest of the turbine structure.

The wind velocity v acting on the structure can be decomposed into two components: the mean velocity $\bar{v}$ and the turbulent velocity $\tilde{v}$, such that $v = \bar{v} + \tilde{v}$. The mean velocity component at height z can be expressed as:

$$\bar{v}(z) = v(z_{\text{ref}}) \left(\frac{z}{z_{\text{ref}}} \right)^\alpha \quad (1)$$

where z_{ref} represents the reference height, $v(z_{\text{ref}})$ is the mean velocity at z_{ref} , and α is the wind shear exponent, which is commonly set to 0.2.

The turbulent velocity component $\tilde{v}$ is characterized by the IEC Kaimal spectral model [36]. According to this model, the Power Spectral Density (PSD) is defined as:

$$S_{\tilde{v}}(f) = \frac{4T I^2 L_K \bar{v}_{\text{hub}}}{(1 + 6f L_K / \bar{v}_{\text{hub}})^{5/3}} \quad (2)$$

where f is frequency in Hz, $\bar{v}_{\text{hub}}$ is the mean velocity at the hub height, TI is turbulence intensity, which can be calculated by:

$$TI = I_{\text{ref}} (0.75 \cdot \bar{v}_{\text{hub}} + 5.6) / \bar{v}_{\text{hub}} \quad (3)$$

where I_{ref} represents the reference turbulence intensity, which can be categorized into four classes: A+, A, B, and C [37]. The integral scale parameter is defined as $L_K = 8.10\Lambda_U$, where the turbulence scale parameter Λ_U is defined as:

$$\Lambda_U = \begin{cases} 0.7H_{\text{hub}}, & H_{\text{hub}} < 60 \text{ m} \\ 0.7 \times 60, & H_{\text{hub}} \geq 60 \text{ m} \end{cases} \quad (4)$$

where H_{hub} is the hub height.

Furthermore, to account for the interaction effects between different spatial points on the turbulent wind speed, the cross spectrum between any two points i and j in space is defined as:

$$S_{ij}(f) = \text{Coh}(i, j, f) \sqrt{S_{ii}(f)S_{jj}(f)} \quad (5)$$

where $S_{ij}(f)$ denotes the cross-PSD between points i and j , while $S_{ii}(f)$ and $S_{jj}(f)$ represent the auto-PSDs for points i and j , respectively.

According to the Kaimal spectral model specified in IEC standard [37], the spatial coherence function is given by:

$$\text{Coh}(i, j, f) = \exp\left(-a \sqrt{\left(\frac{fL}{v_{\text{hub}}}\right)^2 + \left(\frac{0.12L}{L_c}\right)^2}\right) \quad (6)$$

where L is the distance between points i and j , a is the coherence decrement, and L_c is the coherence scale parameter. According to the IEC 61400-1 standard, a is set to 12, and L_c is defined as $8.10\Lambda_U$.

Utilizing Eqs. (1) through (6), a three-dimensional random wind field, which encompasses both the wind turbine's rotor and tower, can be generated accordingly.

3. Co-simulation method for wind-RNA-tower integration

After completing the modeling of the subsystems, this section introduces a comprehensive approach for the integrated coupling of wind, RNA, and tower subsystems, accompanied by verification to ensure model accuracy.

3.1. Description of SCM

For an integrated system comprising MBD and nonlinear FE subsystems, the SCM is a valuable computational approach for handling the complexities of nonlinearity. This method allows for the simulation of each subsystem, representing various physical domains such as mechanics, aerodynamics, hydraulics, or control systems [38].

Given the diverse nature and complexity of these subsystems, modeling the overall system using a single simulation tool or a monolithic approach is often impractical. Instead, each subsystem is independently modeled using specialized software designed for its specific domain. These individual models are then integrated through a co-simulation framework, which manages the data exchange and synchronization between the different models. This flexible approach supports detailed modeling of each subsystem while ensuring their effective interaction within the overall system [39].

As illustrated in Fig. 7(a), the subsystems are extracted from the integrated system. The environmental loads $\mathbf{Q}$ are divided into two components: the loads on the subsystem 1, $\mathbf{Q}_1$, and the loads on the subsystem 2, $\mathbf{Q}_2$. At the cut-interface nodes, which serve as the interaction points between the subsystems, the generalized forces $\mathbf{F}_1$ and $\mathbf{F}_2$ can be derived from the dynamic equilibrium equations of the global model. As shown in Fig. 7(b), the dynamic equations for the subsystem 1 can be expressed as:

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{F}_1 + \mathbf{Q}_1 \quad (7)$$

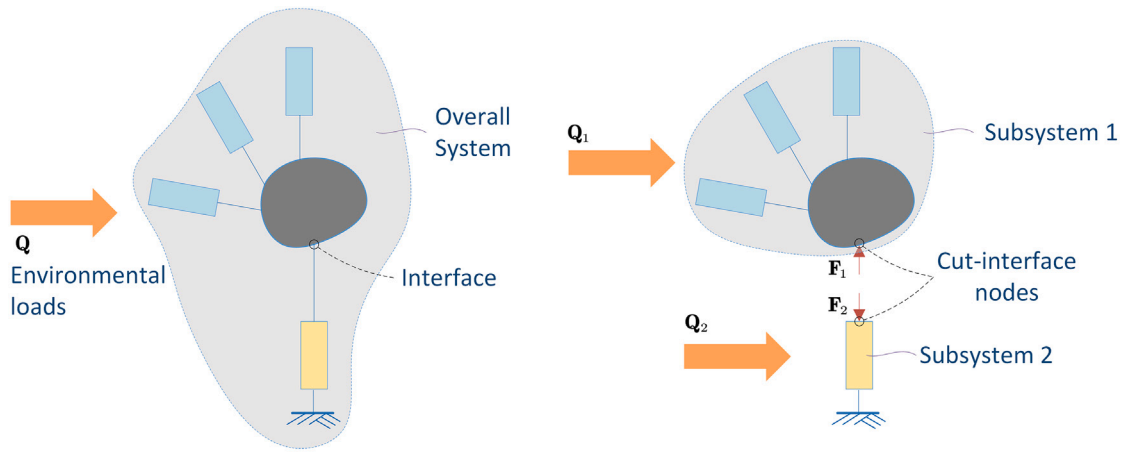
where $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ represent the mass, damping, and stiffness matrices, respectively, $\mathbf{F}_1$ and $\mathbf{Q}_1$ are vectors of generalized forces and loads, respectively, and the generalized coordinate vector is denoted by $\mathbf{x}$.

The dynamic response of the subsystem 2 can be determined by solving for it using the explicit or implicit numerical time integration methods $g(\cdot)$ [26]. The subsystems are processed using separate solvers. The Co-Simulation Engine (CSE) collects the response variables, such as displacement, velocity, or acceleration, at each time step from both subsystems. It then calculates the generalized forces required for the next iteration of the simulation [40].

Based on the introduction above, the wind turbine system can likewise be decoupled into two subsystems: the RNA and the tower. The corresponding generalized forces and environmental loads are denoted as $\mathbf{F}_{\text{RNA}}$, $\mathbf{F}_{\text{tower}}$ and $\mathbf{Q}_{\text{RNA}}$, $\mathbf{Q}_{\text{tower}}$, respectively. According to Eq. (7), for a wind turbine, the equation of motion can be expressed as the following form:

$$\begin{bmatrix} \mathbf{M}_{\text{RNA}} & \mathbf{M}_{\text{side,up}} \\ \mathbf{M}_{\text{side,down}} & \mathbf{M}_{\text{tower}} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{x}}_{\text{blade}} \\ \ddot{\mathbf{x}}_{\text{tower}} \end{bmatrix} + \begin{bmatrix} \mathbf{C}_{\text{RNA}} & \mathbf{C}_{\text{side,up}} \\ \mathbf{C}_{\text{side,down}} & \mathbf{C}_{\text{tower}} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{x}}_{\text{RNA}} \\ \dot{\mathbf{x}}_{\text{tower}} \end{bmatrix} + \begin{bmatrix} \mathbf{K}_{\text{RNA}} & \mathbf{K}_{\text{side,up}} \\ \mathbf{K}_{\text{side,down}} & \mathbf{K}_{\text{tower}} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\text{RNA}} \\ \mathbf{x}_{\text{tower}} \end{bmatrix} = \begin{bmatrix} \mathbf{Q}_{\text{RNA}} \\ \mathbf{Q}_{\text{tower}} \end{bmatrix} \quad (8)$$

where the subscript RNA denotes vectors or matrices associated with the Rotor Nacelle Assembly, while the subscript tower refers to vectors or matrices related to the tower.



(a) Diagram of the SCM.



(b) Data exchanges through the co-simulation engine.

Fig. 7. Subsystem decoupling and co-simulation.

To simulate constrained systems, constraint relaxation can be applied, treating constraints as special forces [39]. Based on this theory the cut-interface generalized forces between the RNA and tower can be decoupled and calculated as follows:

$$\mathbf{F}_{\text{RNA}} = -\mathbf{M}_{\text{side,up}} \cdot \dot{\mathbf{x}}_{\text{tower}} - \mathbf{C}_{\text{side,up}} \cdot \dot{\mathbf{x}}_{\text{tower}} - \mathbf{K}_{\text{side,up}} \cdot \mathbf{x}_{\text{tower}} \quad (9)$$

$$\mathbf{F}_{\text{tower}} = -\mathbf{M}_{\text{side,down}} \cdot \dot{\mathbf{x}}_{\text{RNA}} - \mathbf{C}_{\text{side,down}} \cdot \dot{\mathbf{x}}_{\text{RNA}} - \mathbf{K}_{\text{side,down}} \cdot \mathbf{x}_{\text{RNA}} \quad (10)$$

where $\mathbf{F}_{\text{RNA}}$ represents the forces exerted on the RNA due to the tower's response, and $\mathbf{F}_{\text{tower}}$ denotes the forces applied to the tower as a result of the RNA's response. By substituting Eqs. (9) and (10) into Eq. (8), the equation of motion can be derived as follows:

$$\mathbf{M}_{\text{RNA}} \cdot \ddot{\mathbf{x}}_{\text{RNA}} + \mathbf{C}_{\text{RNA}} \cdot \dot{\mathbf{x}}_{\text{RNA}} + \mathbf{K}_{\text{RNA}} \cdot \mathbf{x}_{\text{RNA}} = \mathbf{F}_{\text{RNA}} + \mathbf{Q}_{\text{RNA}} \quad (11)$$

$$\mathbf{x}_{\text{tower}}, \dot{\mathbf{x}}_{\text{tower}}, \ddot{\mathbf{x}}_{\text{tower}} = g(\mathbf{F}_{\text{tower}} + \mathbf{Q}_{\text{tower}}) \quad (12)$$

By decoupling the system, the subsystems can be effectively isolated and analyzed independently. The velocity, displacement, and force at the cut-interface nodes are used as coupling variables [26]. In each analysis sub-step, Simpack solves the motion of the RNA system and transmits the conjugate reaction force to Abaqus. Abaqus then performs the nonlinear FEA of the tower and returns the displacement and velocity at the cut-interface node to Simpack [41]. Force coupling enforces action–reaction balance at cut-interface nodes, theoretically reducing unbalanced forces and enhancing simulation stability. Displacement and velocity coupling maintains consistent motion states, preventing error accumulation from movement inconsistencies and thus improving simulation accuracy. Assume that at time step i , the RNA subsystem's dynamic response vectors $\mathbf{x}_{\text{RNA}}^i$, $\dot{\mathbf{x}}_{\text{RNA}}^i$ and $\ddot{\mathbf{x}}_{\text{RNA}}^i$ are known. Solving equation (11) then yields the force vector $\mathbf{F}_{\text{RNA}}$ at time step $i + 1$. According to Newton's Third Law, action and reaction forces are equal in magnitude and opposite in direction:

$$\mathbf{F}_{\text{tower}}^{i+1} = -\mathbf{F}_{\text{RNA}}^{i+1} \quad (13)$$

Apply $\mathbf{F}_{\text{tower}}^{i+1}$ as a boundary load to the tower model, then perform the FE computation as per equation (12). To maintain deformation compatibility between the two subsystems and ensure motion consistency at the interface, Abaqus transmits the motion state computed at time step $i + 1$ to Simpack by feeding back the velocity and displacement of the cut-interface nodes. In Simpack, this boundary information is introduced as a constraint in the system's equations of motion. Numerical integration is then performed to obtain the dynamic response vectors of the RNA subsystem, $\mathbf{x}_{\text{RNA}}^{i+1}$, $\dot{\mathbf{x}}_{\text{RNA}}^{i+1}$, $\ddot{\mathbf{x}}_{\text{RNA}}^{i+1}$, at time step $i + 1$. This process repeats until the analysis is complete.

The cut-interface node of the RNA subsystem is defined at the central point of the yaw flange. For the SCHT subsystem, the cut-interface node is positioned at the centroid of the tower top. This node is then kinematically coupled to the surrounding FE nodes at the tower top, establishing a connection that allows for the transfer of motion [41]. After establishing the subsystems as described in Section 2, the cut-interface nodes of the RNA and SCHT are associated. Data exchange at the cut-interface is handled

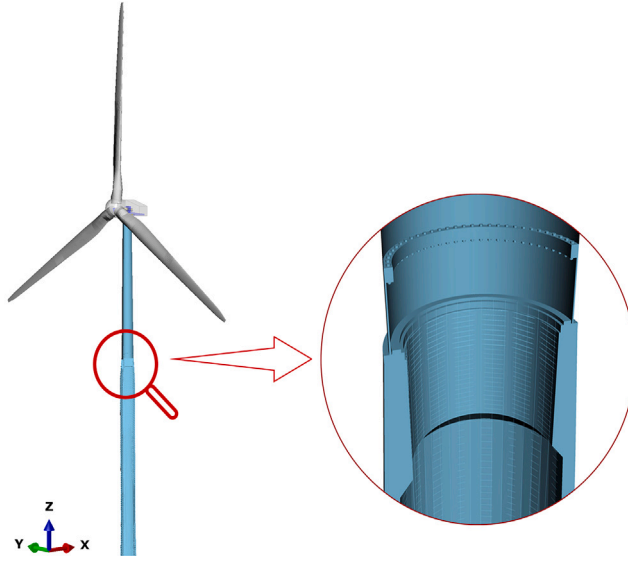


Fig. 8. The MBD-FE model of the wind turbine with SCHAT.

by the CSE integrated within Simpack. This engine is designed to manage the intricate process of data interchange between the coupled subsystems, ensuring seamless integration and coordinated simulation of the overall system. The global coordinate system's orientation is consistent with the local coordinate system of the tower, with the Y-axis lying in the plane of the rotor and the Z-axis pointing towards the top of the tower. The final MBD-FE model of the wind turbine is illustrated in Fig. 8.

3.2. Aerodynamic interaction

The aerodynamic loads on the structure of a wind turbine include blade aerodynamic loads and tower aerodynamic loads. Simpack also allows for the complex interactions between components, including the effects of external forces like wind. By inputting the turbulent wind field files into Simpack's AeroDyn force element, the aerodynamic loads can be applied. The methods commonly used to apply aerodynamic forces to blades are usually the Blade Element Momentum (BEM) or the Generalized Dynamic Wake (GDW) theory [42]. The aerodynamic loads are determined employing the BEM theory, chosen for its computational efficiency and high accuracy compared to computational fluid dynamics [43]. The BEM theory is an amalgamation of the blade element theory and momentum theory. It involves calculating the normal and tangential force time history acting on each blade element through the momentum theory, taking into account the aerodynamic characteristics of the blade and the time history of the incoming wind speed [44,45]. As illustrated in Fig. 9, each blade is discretized into mutually independent blade elements. In each airfoil unit, consistent aerodynamic properties such as chord length, helix angle, and angle of attack are assumed in the length direction. This approach facilitates a detailed and effective analysis of the aerodynamic behavior of the wind turbine's blades.

Considering the relative wind speed in the normal direction $v(1-a)$, and the relative wind speed in the tangential direction $\Omega r(1+a')$ due to the angular rotation velocity Ωr of the blades, the resultant relative velocity of the blade element can be expressed as:

$$v_{\text{rel}} = \sqrt{[v(1-a)]^2 + [\Omega r(1+a')]^2} \quad (14)$$

where a and a' are the axial and tangential induction factors that can be calculated by iterative computations. v is the turbulent wind velocity simulated using the method described in Section 2.3.

The flow angle ϕ , which is defined as the angle between the inflow direction v_{rel} and the plane of rotation of the blades, can be calculated by the following Equation [43]:

$$\phi = \tan^{-1} \frac{v(1-a)}{\Omega r(1+a')} \quad (15)$$

The local attack angle, denoted as α , is defined as the angle between the rotation plane and the chord line. The formula is as follows:

$$\alpha = \phi - \theta \quad (16)$$

where θ is the angle that results from the combination of the pitch angle and the twist.

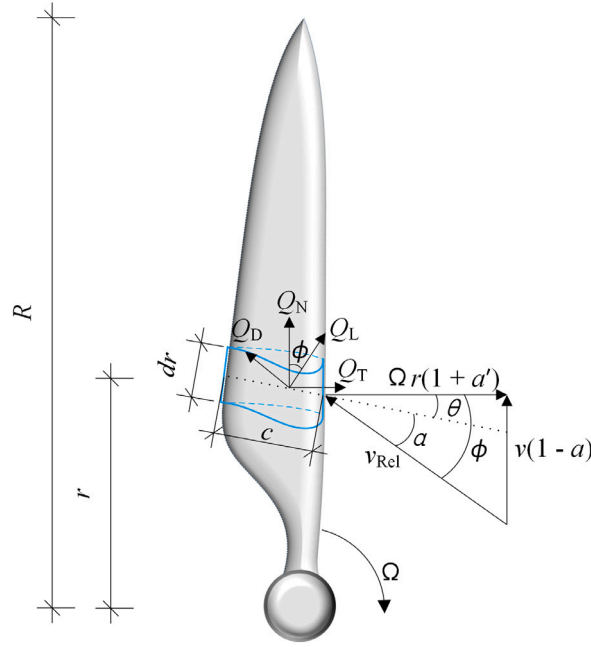


Fig. 9. Schematic diagram of blade element.

Once the attack angle is predetermined, the lift and drag coefficients C_L and C_D can be derived by referencing the airfoil data. Subsequently, the lift force Q_L and drag force Q_D per unit length can be expressed as:

$$\begin{aligned} Q_L &= \frac{1}{2} \rho v_{rel}^2 c C_L \\ Q_D &= \frac{1}{2} \rho v_{rel}^2 c C_D \end{aligned} \quad (17)$$

where ρ is the density of air, which can take the value of $\rho = 1.23 \text{ kg/m}^3$, c is the length of the chord.

The normal coefficients C_N and tangential coefficients C_T are introduced as:

$$\begin{aligned} C_N &= \cos \phi C_L + \sin \phi C_D \\ C_T &= \sin \phi C_L - \cos \phi C_D \end{aligned} \quad (18)$$

Further, the forces Q_N and Q_T , which is respectively normal to and tangential to the rotation plane, can be calculated by decomposing Q_L and Q_D along the two directions:

$$\begin{aligned} Q_N &= \frac{1}{2} \rho v_{rel}^2 c C_N \\ Q_T &= \frac{1}{2} \rho v_{rel}^2 c C_T \end{aligned} \quad (19)$$

For the S_{CHT} structure, aerodynamic loads can be calculated using the quasi-steady assumption. The aerodynamic loads per unit height z of the tower can be expressed as [5,46]:

$$dQ_{tower} = \frac{1}{2} C_{d,t} D_t \rho (v - v_t)^2 dz \quad (20)$$

where $C_{d,t}$ and D_t are the drag coefficient and outer diameter of the tower, respectively, and v_t is unit velocity of the tower.

3.3. Method verification

To verify the applicability of SCM for dynamic analysis of wind turbines, a numerical model of the NREL 5 MW reference wind turbine is implemented using both the wind-RNA-tower integrated coupling program and the OpenFAST program [47]. The tower's material properties are specified as follows: a density of 8500 kg/m^3 , Young's modulus of $2.1 \times 10^{11} \text{ Pa}$, and a Poisson's ratio of 0.3. The turbine model features a hub height of 90.0 m, a rotor diameter of 126.0 m, and a rated rotational speed of 12.1 Revolutions Per Minute (RPM). For wind conditions, the Normal Turbulence Model (NTM) is employed, with turbulence intensity classified as A. The average wind speed at the hub height is set to 14 m/s.

The comparative results are presented in Fig. 10, and the deformation diagrams of the MBD-FE model are shown in Fig. 11. These results demonstrate that the dynamic response from the SCM-based model are largely consistent with those obtained from OpenFAST. In the initial response phase, the curves reveal that the SCM offers enhanced stability compared to OpenFAST, which

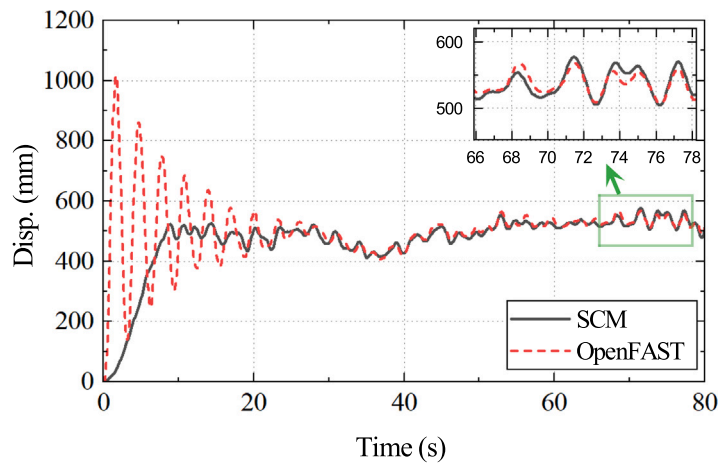
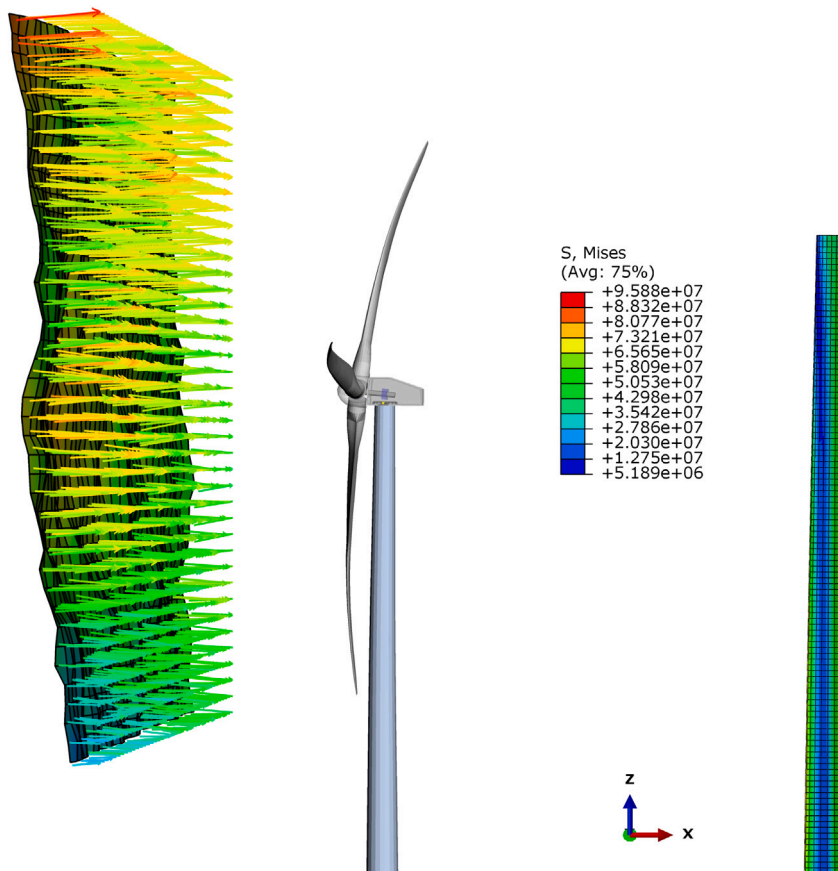
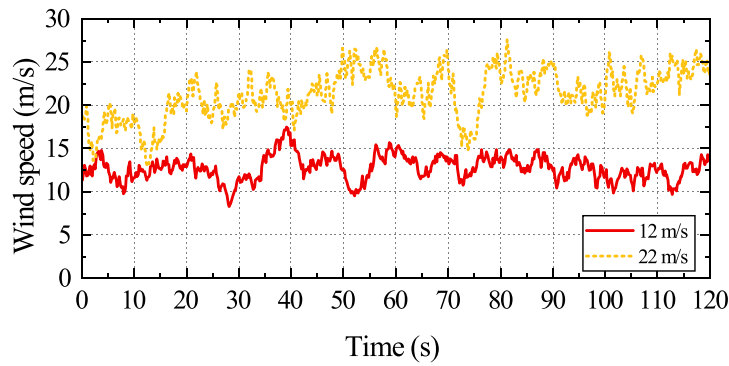


Fig. 10. Displacement comparison at the tower top of the reference wind turbine.

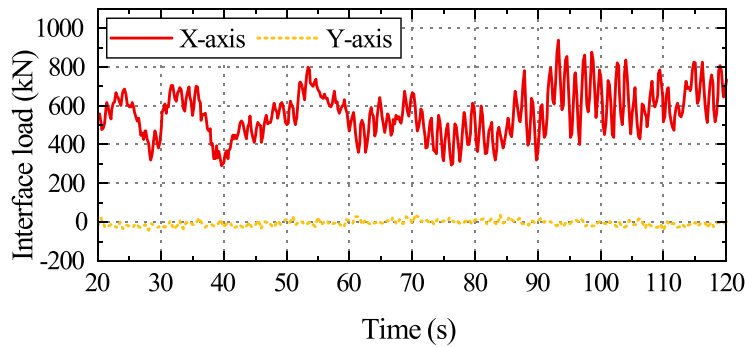


(a) The integrated coupling model. (b) Stress contour map of the tower.

Fig. 11. Deformation diagrams.



(a) Wind velocity at the hub height.



(b) Wind loads at tower top.

Fig. 12. Wind velocity and loads.

helps avoid premature plastic deformation or damage to the material during subsequent nonlinear analysis [48,49]. After 20 s of wind loads application, the tower's dynamic response stabilizes with minimal fluctuations. Overall, the integrated coupling model created using SCM shows excellent analytical accuracy and is well-suited for detailed SHT analysis.

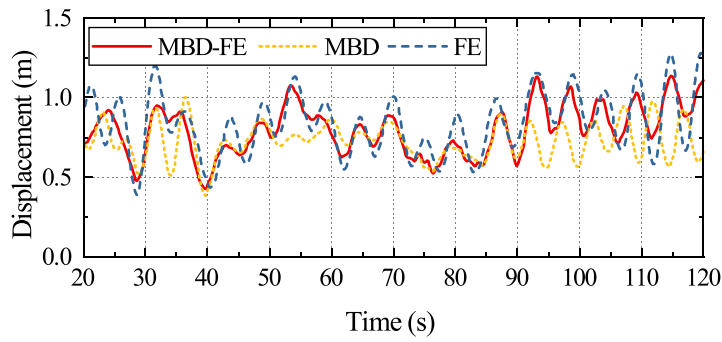
4. Results and discussions

The proposed SCM enables detailed modeling of the SHT structure introduced in Section 2.2. In the following sections, dynamic analysis of the SHT wind turbine with the integrated coupling model will be performed under rated and high wind speeds.

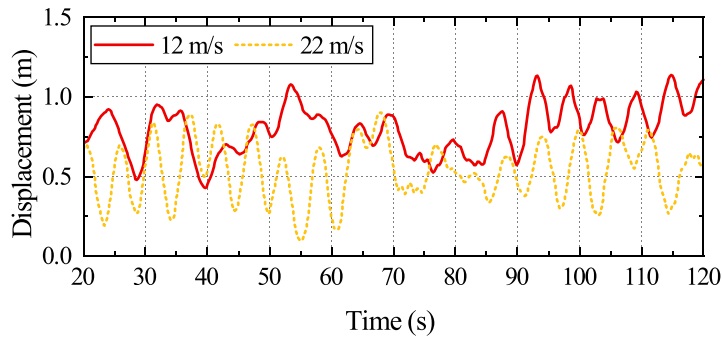
4.1. Integrated dynamic analysis

As described in Section 2.3, a three-dimensional wind field is generated with a hub height of 160 m, where the reference turbulence intensity I_{ref} is classified as class A. Fig. 12(a) displays representative time history curves over 120 s for wind velocity at the hub height. The curves consider scenarios with an average hub wind speed of $\bar{v}_{hub} = 12$ m/s and turbulence intensity $TI = 19.5\%$, as well as an average hub wind speed of $\bar{v}_{hub} = 22$ m/s and turbulence intensity $TI = 16.1\%$. To eliminate the impact of transient response on dynamic analysis results, it is common to discard the initial response period of the analysis [50]. The analysis excludes the initial 20 s to avoid the transient instability commonly observed when the wind turbine starts. Therefore, the data are analyzed over a period of 100 s. Fig. 12(b) illustrates the time history of the wind loads at the cut-interface node for a hub wind speed of $\bar{v}_{hub} = 12$ m/s. The results indicate that the loads applied to the tower along the X-axis are significantly higher compared to those along the Y-axis. Consequently, the focus of the following analyses will be predominantly on the performance and dynamics in the X-axis direction.

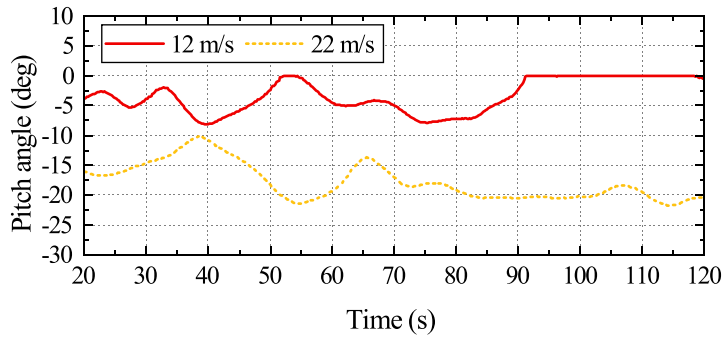
As shown in Fig. 13(a), the responses of the MBD-FE model, MBD model, and FE model under a wind speed of 12 m/s are compared. The MBD-FE model, a coupled nonlinear model developed using SCM, accounts for geometric, material, and boundary nonlinearities. The MBD model is a coupled linear model built with OpenFAST, while the FE model is an uncoupled nonlinear model, in which the wind loads is extracted from OpenFAST and applied to the tower [51]. The results indicate that the overall displacement trends at the tower top are consistent across all three models. Due to the lack of consideration for the tower's nonlinear



(a) Displacement curves of three types of models.

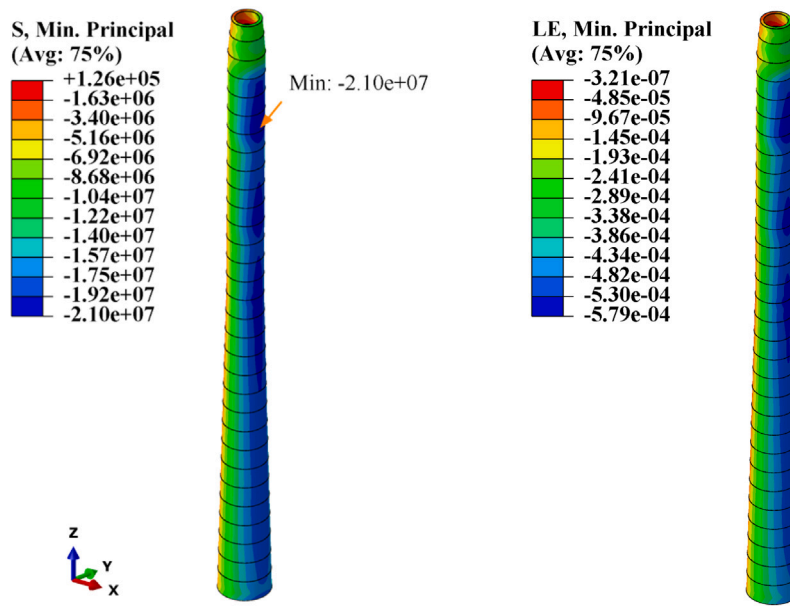


(b) Displacement curves of MBD-FE model.

Fig. 13. Displacement at tower top.**Fig. 14.** Pitch angle.

characteristics, the MBD model exhibits smaller displacement at the tower top. While the FE model incorporates the tower's nonlinear features, it neglects aerodynamic coupling, and the absence of aerodynamic damping leads to larger dynamic response. Therefore, the MBD-FE model, which simultaneously considers both nonlinearities and integrated coupling effects, is essential for accurate wind turbine dynamic simulations.

Subsequently, the impact of different wind speeds on the MBD-FE model is compared. As depicted in Fig. 13(b), despite the lower average wind speed of 12 m/s, the top displacement of the tower is notably higher. The increased displacement is due to the blade pitch angle being nearly 0 degrees near the rated wind speed, as shown in Fig. 14, resulting in greater aerodynamic loads from the wind. Therefore, effective blade pitch control is crucial for managing the tower's displacement. At an average wind speed of 12 m/s, the tower experiences a maximum displacement of 1.138 m. Fig. 15 illustrates that under this displacement, the maximum compressive stress in the concrete occurs on the leeward side of the 26th section of the tower. Here, the compressive stress reaches 21.0 MPa with a corresponding true strain of 0.0579%. This area is characterized by a reduction in the wall thickness of the concrete



(a) Stress contour map.

(b) Strain contour map.

Fig. 15. Stress and strain contour map of SCHT.

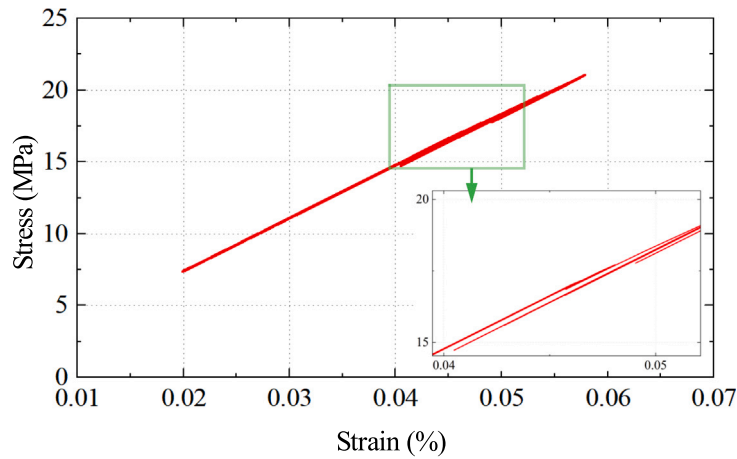


Fig. 16. Strain-stress curve at the location of maximum stress.

segments and a decrease in the outer diameter. Fig. 16 presents the strain-stress curve at the point of maximum stress, indicating that as the load increases, the concrete begins to exhibit plastic deformation. Consequently, the material's stiffness decreases over time, leading to higher strain levels for the same stress. Thus, in the structural design of the tower, it is essential to pay close attention to stress states at locations where the outer diameter and wall thickness are reduced. Fig. 17 displays the stress contour map for the steel structural components at maximum displacement. The figure reveals that, unlike the concrete structure, the transition section, reinforcing rebars, and PT tendons are effectively stressed and have not entered the plastic strain region. Stress concentration is evident at the RF connection, with the bolts on the windward side experiencing peak stresses up to 755 MPa. This concentration of stress is attributed to the stiffness discontinuity at the transition section where the concrete and steel tower segments connect. In the tower's design, particular attention must be given to potential bolt failure at the RF connection, as highlighted by the stress concentration observations [52,53].

Through the above analysis, it is evident that the adoption of the SCM is crucial in capturing these complex interactions. Compared to existing MBD methods such as OpenFAST, the integrated MBD-FE approach offers a more accurate simulation of stress and strain distribution, particularly in critical components, enabling a more reliable assessment of potential failure points. A key

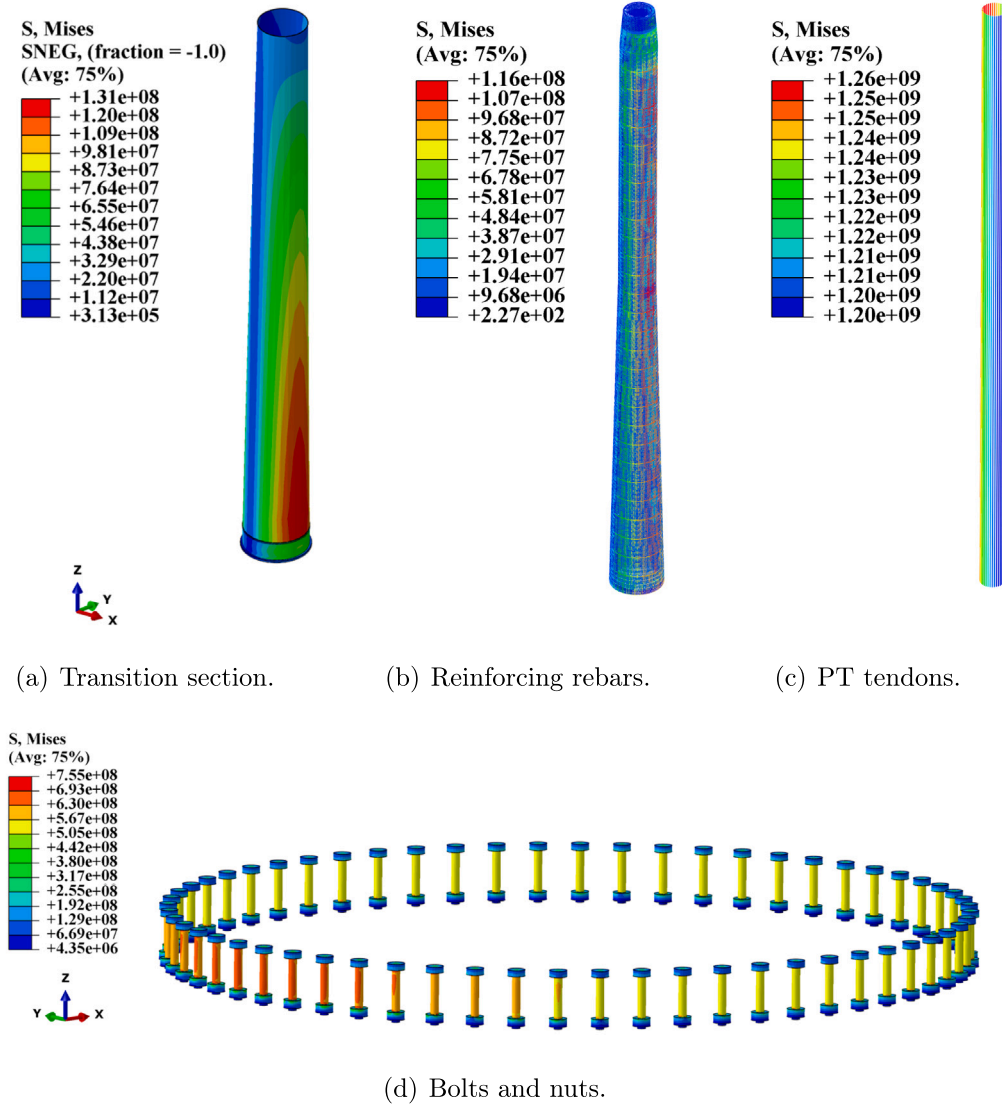


Fig. 17. Stress contour map of steel components.

advantage of the proposed SCM over OpenFAST is its ability to manage the complex, nonlinear dynamic behavior of wind turbines, especially those with SCHT. The SCM integrates MBD and FE models, allowing for accurate simulation of nonlinear behaviors and nonlinear time history analyses, essential for capturing dynamic responses under varying wind loads and complex conditions. Additionally, the high-fidelity dynamic analysis of the proposed method couples detailed FE modeling of the SCHT with MBD simulations for the RNA, providing precise modeling of the tower structure, including key components like PT tendons and RF connections. In contrast, OpenFAST prioritizes computational efficiency but is limited in the structural detail it can capture. In contrast to traditional FEA methods, the integrated approach can realistically simulate the coupling effect between environmental loads and the structure, while also taking into account the operational state of the wind turbine control system. Therefore, the SCM offers significant advantages by enabling the simultaneous simulation of both subsystems, enhancing the accuracy of numerical analysis.

4.2. A novel transition section design

The dynamic analyses above reveal significant stress on the bolts at the RF connection within the transition section. To mitigate this issue, this section introduces an innovative transition section design aimed at enhancing stress distribution across the bolts.

The proposed novel transition section structure, illustrated in Fig. 18, maintains the same interface dimensions as the existing concrete and steel tower sections. The new design preserves the dimensions of the upper and lower RFs, each with a thickness

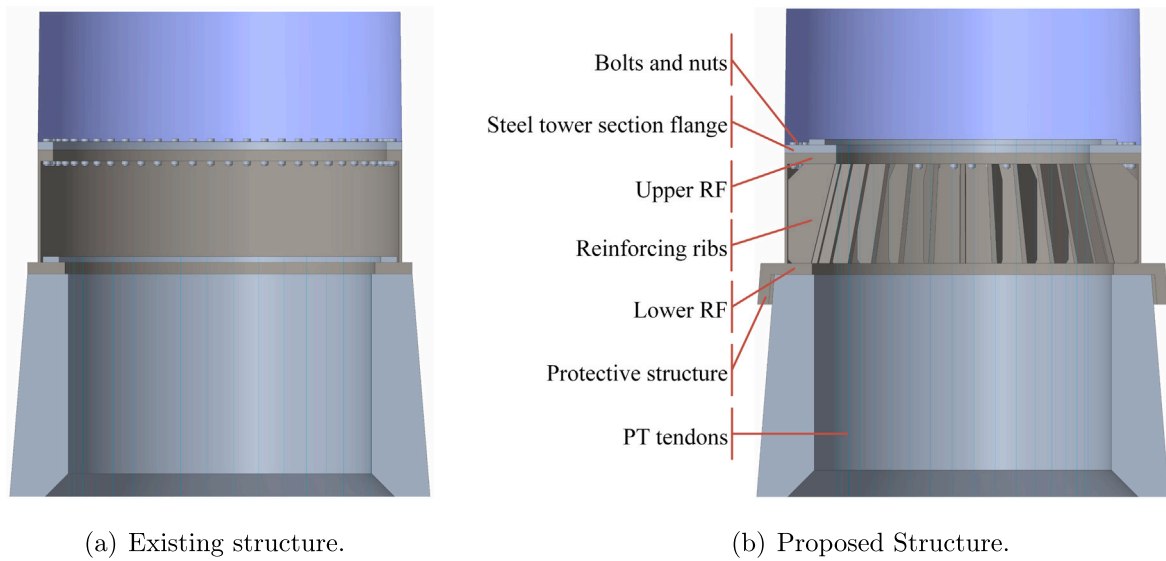


Fig. 18. Transition section structures.

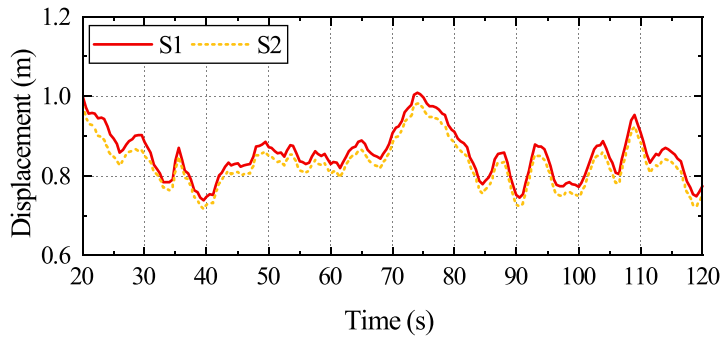


Fig. 19. Displacement at tower top.

of 140.0 mm, and the tube wall thickness of 35.0 mm, consistent with the current structure. Additionally, the bolt diameter and quantity remain unchanged. The key features of the new and existing structures are described as follows:

- (1) Existing structure (S1): This configuration features a straightforward design, comprising upper and lower RFs along with an intermediate steel tube. PT tendons are anchored to the lower flange.
- (2) Proposed structure (S2): Reinforcing ribs are introduced between the upper and lower RFs, with bolted RF connections and PT tendons securing both the steel tower section flange and the upper flange simultaneously. Reinforcing ribs are incorporated between the upper and lower RFs, with bolted RF connections and PT tendons securing both the steel tower section flange and the upper flange. Based on engineering experience, this design is effective in enhancing the stiffness at the base of the steel tower section. To address issues related to low friction between the steel and concrete, and to mitigate the risk of partial separation at the contact surfaces, a protective structure is added around the perimeter of the lower flange. This addition prevents slippage between the transition section and the concrete section.

Comparative time history analyses with both configurations, illustrated in Fig. 19, show that the tower top displacement in the X-axis direction stabilize around 0.9 m. The maximum displacement for both configurations are detailed in Table 3. The results indicate that the maximum displacement for the transition section with reinforcing ribs (structure S2) is notably lower than that for the standard transition section (structure S1). This demonstrates that incorporating reinforcing ribs enhances the stability and reduces displacement of the tower.

Fig. 20 illustrates the maximum stress experienced by bolts, which is observed to be highest at the outer side of the tower in the bending direction. Figs. 21 and 22 display the time history curves for the bolts and PT tendons under maximum stress conditions. It is evident from these figures that the bolt stresses in the S1 structure are significantly higher compared to those in the S2 structure. Table 3 quantifies this difference, showing that the stress amplitudes for bolts are 143.16 MPa for S1 and 80.51 MPa for S2. This

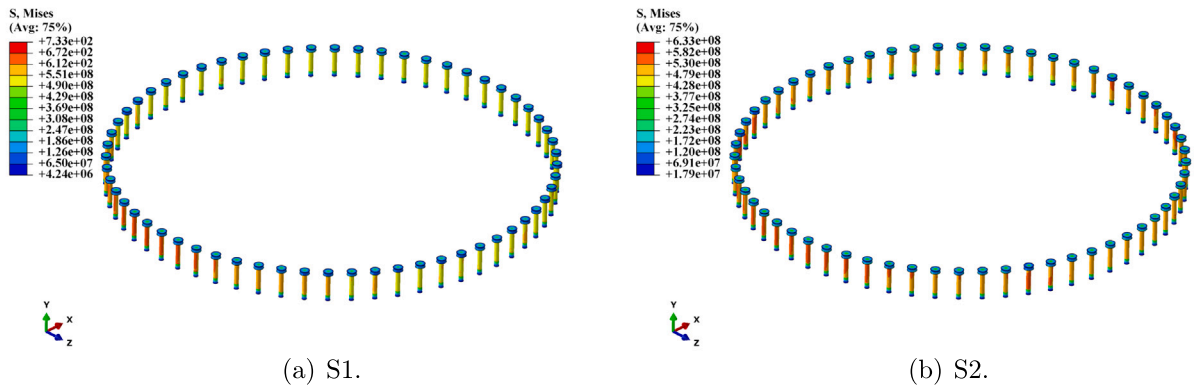


Fig. 20. Stress of bolts.

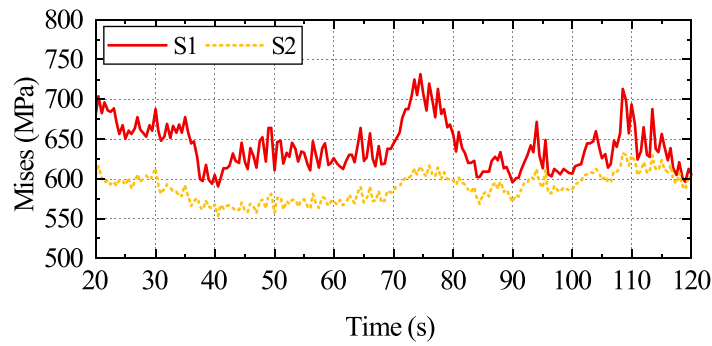


Fig. 21. Stress of bolts.

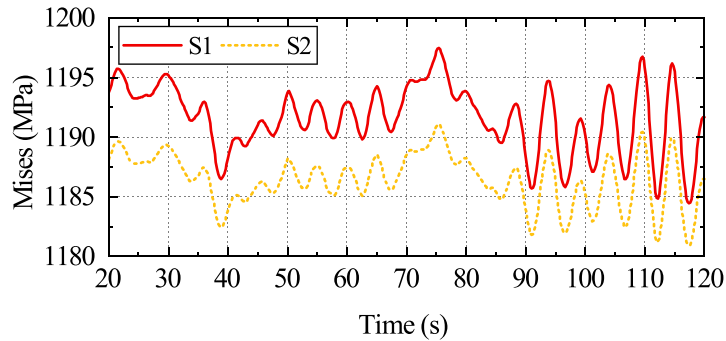


Fig. 22. Stress of PT tendons.

indicates that the S2 structure not only maintains lower mean and amplitude stress levels for bolts but also preserves the stress levels on PT tendons. According to the Palmgren-Miner linear damage accumulation criterion [54,55], the reduction in bolt stress amplitude is likely to extend their service life [53,56], thereby enhancing the overall reliability of the bolted RF connections.

5. Conclusion

This study proposes a high-fidelity MBD-FE SCM to conduct nonlinear time history analysis of onshore wind turbines with SHT. By integrating the well-established commercial tools Simpack for MBD and Abaqus for FEA, the proposed SCM facilitates

Table 3
Comparisons of displacement and stress.

Structure	Max displacement (m)	Bolts			PT tendons		
		Min stress (MPa)	Max stress (MPa)	Stress amplitude (MPa)	Min stress (MPa)	Max stress (MPa)	Stress amplitude (MPa)
S1	1.086	590.29	733.45	143.16	1184.43	1197.46	13.03
S2	1.057	553.37	633.88	80.51	1180.90	1191.05	10.15

the establishment of a comprehensive and efficient MBD-FE model. This method showcases robust adaptability, efficiently meeting diverse modeling needs. The dynamic response analyses performed assess the effects of varying wind speeds on the entire wind turbine system, including the bolted RF connections and PT tendons. The principal conclusions of this study are as follows:

- (1) Compared to the results obtained from OpenFAST, the MBD-FE model developed using the SCM demonstrates both high accuracy and stability in its analyses. This model allows for comprehensive and detailed representation of the tower structure while performing nonlinear time history analysis. Unlike traditional dynamic models, the MBD-FE approach integrates detailed modeling of the tower and accounts for the coupling effects of the wind field. This capability enables a more precise capture of the structural response under complex aerodynamic loads, enhancing both simulation accuracy and reliability. Such improvements are crucial for the design and optimization of wind power towers, offering valuable insights for more effective engineering solutions.
- (2) Dynamic analyses reveal that blade pitch control significantly mitigates tower displacement. When the blade pitch angle is set to 0 degrees, the wind turbine encounters maximum aerodynamic loads. The steel tower section demonstrates superior stress performance compared to the concrete section, where the concrete structure experiences plastic deformation. Consequently, when designing the SCHAT structure, particular attention should be given to two critical areas: first, the regions where the concrete segment's outer diameter and wall thickness reduce, and second, the RF connection at the transition section.
- (3) Under wind load conditions, the SCHAT incorporating the S2 transition section shows a marked decrease in the top tower displacement compared to the S1 configuration. This improvement highlights the effectiveness of the reinforcing ribs in enhancing the structural stiffness at the base of the steel tower section. Additionally, the design utilizing bolted RF connections and PT tendons to secure both the steel tower flange and the upper flange significantly reduces both the average and maximum bolt stresses. This modification not only improves the durability of the bolts but also maintains the reliability of the bolted RF connection, without compromising the performance of the PT tendons.

The intricate stress conditions identified in the SCHAT structure highlight the need for extensive engineering validation to ensure its long-term reliability. Future studies will concentrate on a deeper examination of these stress conditions. Additionally, further theoretical development is required to optimize complex FE models and improve analysis efficiency.

Replication of results

Some or all data, models, or code generated or used during the study are available from the corresponding author by request.

CRediT authorship contribution statement

Dan Wang: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation. **Jun Xu:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Zeyu He:** Writing – review & editing, Software. **Quanfu Yu:** Writing – review & editing, Methodology. **Weiwei Tang:** Writing – review & editing, Methodology. **Guangling He:** Funding acquisition, Conceptualization. **Qiang Wu:** Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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